

A REPORT

ON

**GAN and Diffusion Based Framework for Robust
Deepfake Detection Across Image Modalities**

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Abstract: The rapid progress of generative models such as GANs and diffusion models has made synthetic and manipulated face images nearly indistinguishable from real ones, creating a pressing need for reliable detection. This project develops a robust, explainable framework for distinguishing real from fake facial images. A Vision Transformer baseline fine-tuned on a single-generator StyleGAN dataset achieves near-perfect in-domain accuracy, but a controlled cross-dataset study on the Diverse Fake Face Dataset (DFFD) shows that detection of unseen manipulation types collapses to roughly chance-level recall, exposing a severe generalization gap. Fine-tuning on diverse fakes recovers performance substantially, and a per-generator analysis isolates subtle attribute edits (FaceApp) as the hardest case. Building on these findings, a three-stream detector – RGB, frequency and noise-residual encoders fused by cross-attention – was implemented and trained end-to-end. On the held-out DFFD test set it reaches a fake-detection recall of 0.992 and ROC-AUC of 0.999, lifting FaceApp recall from 0.45 to 0.93 and raising overall fake recall from the 0.29 baseline to 0.99. The model exposes per-stream confidences and saliency maps for interpretability, and is deployed behind an interactive interface for image-level verification.

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Chapter 1

Introduction

Deepfakes are synthetic or manipulated media in which a person’s face is generated, swapped, or altered using deep learning. Driven by generative adversarial networks (GANs) [6] and, more recently, diffusion models [7], the visual quality of fake faces has reached a point where humans can no longer reliably tell them apart from genuine photographs. While these techniques have many legitimate uses, their misuse fuels misinformation, identity fraud, financial scams and reputational harm. Automatic, trustworthy detection of fake facial imagery has therefore become an important problem in media forensics and AI security.

1.1 Motivation

Two trends make detection difficult. First, the diversity of generators is growing rapidly: entire-face synthesis (StyleGAN [9], PGGAN [8]), attribute and expression manipulation (StarGAN [2], consumer apps such as FaceApp), and diffusion-based generation each leave different traces. Second, detectors trained on one generator often fail to recognise fakes from another, so high reported accuracy on a single benchmark can hide poor real-world reliability. A practical detector must therefore generalize across generators and remain robust to everyday image degradations such as compression and resizing.

1.2 Objectives

The objectives of this project are:

- To develop a robust AI-driven framework for the detection of deepfake and manipulated images produced by GAN- and diffusion-based generative models.
- To design multi-modal forensic techniques capable of identifying synthetic artifacts, inconsistencies and hidden manipulation traces across different image modalities.
- To leverage advanced deep learning architectures, including CNNs, Vision Transformers (ViTs), GAN discriminators and diffusion-aware detection models, for enhanced deepfake identification.
- To create a generalized detection framework that is resilient against evolving generative techniques, adversarial attacks, compression and post-processing operations.

1.3 Scope and Organization

This report covers the full arc of the project: establishing a strong single-model baseline, quantifying its cross-dataset generalization, and then implementing, training and evaluating the proposed multi-encoder architecture end-to-end. Chapter 2 reviews related work. Chapter 3 describes the datasets, the baseline model, the cross-dataset protocol and the multi-stream framework as built. Chapter 4 presents and discusses the baseline, cross-dataset, and final multi-stream results. Chapter 5 covers the interactive verification interface. Chapters 6 and 7 give conclusions, future work and recommendations.

Chapter 2

Literature Review

Generative models. GANs [6] introduced adversarial training between a generator and a discriminator, and progressive and style-based variants [8, 9] brought photorealistic face synthesis. Attribute-translation models such as StarGAN [2] edit specific facial attributes rather than synthesizing a whole face. Diffusion models [7] generate images by iteratively denoising and now rival or exceed GANs in quality, representing the newest class of fake-image sources.

Detection approaches. Early detectors used convolutional neural networks trained as binary classifiers, and CNN-generated images were shown to be detectable across many architectures when trained appropriately [13]. Frequency-domain methods exploit spectral artifacts left by upsampling layers in GANs [5]. Vision Transformers [4] and self-supervised transformer features such as DINO [1] have been applied to forgery detection, using global self-attention to capture subtle, spatially distributed inconsistencies. Noise-residual and steganalysis-derived features, such as the SRM high-pass filters of Zhou et al. [15], expose local tampering by suppressing image content and amplifying sensor-noise inconsistencies.

Benchmarks. Standard datasets include FaceForensics++ [11] and Celeb-DF [10] (video face-swap / reenactment) and the Diverse Fake Face Dataset (DFFD) [3], which gathers real faces alongside fakes from multiple generators (FaceApp, StarGAN, PGGAN, StyleGAN). The 140k Real and Fake Faces collection [14] pairs real FFHQ images with StyleGAN fakes and is a common single-generator starting point.

Generalization and explainability. A recurring finding is that detectors generalize poorly across generators, which motivates diverse training and cross-dataset evaluation. For interpretability, gradient-based methods such as Grad-CAM [12] and transformer attention maps highlight the regions driving a decision, which is essential for trustworthy forensic use.

Chapter 3

Methodology

3.1 Problem Formulation

Deepfake detection is treated as binary image classification. Given a face image x , a model f_θ predicts a class $\hat{y} \in \{\text{real}, \text{fake}\}$ from class probabilities obtained via the softmax function,

$$\hat{y} = \arg \max_c \text{softmax}(f_\theta(x))_c. \quad (3.1)$$

The model is trained by minimising the binary cross-entropy loss over N samples,

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right], \quad (3.2)$$

where p_i is the predicted probability of the fake class. We treat fake as the positive class, so recall measures the fraction of fakes successfully caught.

3.2 Datasets

Two datasets are used. The 140k Real and Fake Faces dataset [14] contains 70,000 real (FFHQ) and 70,000 StyleGAN-generated images, pre-split into training, validation and test sets, and represents a single-generator setting. The Diverse Fake Face Dataset (DFFD) [3] provides real FFHQ faces together with fakes from FaceApp and StarGAN (attribute manipulation) and PGGAN and StyleGAN (entire synthesis). DFFD ships with an official, identity-disjoint train / validation / test split (roughly 50/5/45), which we use as-is: a random per-image split can place the same identity in both training and test partitions and silently inflate recall. All images are resized to 224×224 . Following the multi-stream design (Section 3.5), images are supplied to the network as raw tensors in $[0, 1]$ and each encoder applies its own preprocessing internally.

3.3 Baseline Model

The baseline is a Vision Transformer (ViT-Base/16) [4], pre-trained on ImageNet and fine-tuned as a two-class classifier using transfer learning. Training uses the AdamW optimiser, cross-entropy loss (Eq. 3.2) and mixed-precision computation; the checkpoint with the best validation accuracy is retained. Starting from a pre-trained backbone allows the model to converge in only a few epochs.

3.4 Cross-Dataset Evaluation Protocol

To measure generalization, the baseline (trained only on StyleGAN fakes) is evaluated as-is on the DFFD test set, giving a “before” result. The same model is then fine-tuned on a balanced DFFD training split and re-evaluated on the identical test set, giving an “after” result. Reporting both isolates the effect of exposure to diverse fakes. Performance is summarised by accuracy, precision, recall, F1 and ROC-AUC, plus a per-generator detection rate.

3.5 Multi-Encoder Framework

Single-stream models capture only one view of an image. The framework, shown in Fig. 3.1, processes an input image through three parallel encoders: an **RGB encoder** (ViT-Base/16, initialised from

the DFFD fine-tuned baseline) for visual semantics; a **frequency encoder** that computes the two-dimensional FFT log-magnitude spectrum and passes it through a compact CNN, targeting spectral artifacts from generator upsampling [5]; and a **noise-residual encoder** that applies fixed SRM high-pass filters [15] together with a learnable Bayar constrained convolution before a CNN, targeting local noise and compression inconsistency. Each encoder emits a sequence of tokens.

The three token sequences are combined by a **cross-attention fusion** transformer. A learnable stream-type embedding tags every token, a learnable fusion token is prepended, and self-attention over the concatenated sequence lets each stream attend to every other; conceptually each attention step computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V. \quad (3.3)$$

The fusion token is read by a **deepfake classifier** that outputs a real/fake score. An auxiliary classification head is kept on each stream during training (deep supervision), which prevents the strong pretrained RGB stream from suppressing the two freshly initialised streams. The model additionally exposes a saliency-based explainability map (Chapter 5).

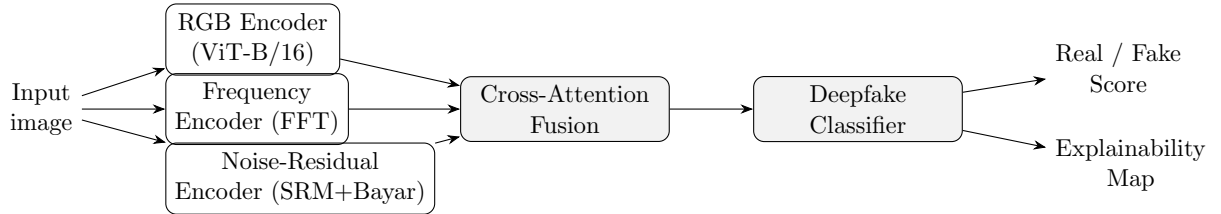


Figure 3.1: Multi-encoder deepfake detector: three parallel encoders fused by cross-attention, producing a real/fake score and an explainability map.

3.6 Training Strategy

The fusion model is trained in stages to keep every stream informative. First, the frequency and noise streams are warm-started individually as real/fake classifiers. Second, the fusion module is trained with the RGB backbone frozen. Empirically (Section 4.4), keeping the backbone frozen throughout gave the best and most stable result, so end-to-end unfreezing was dropped from the final recipe. The total loss is the cross-entropy of the fusion classifier plus a weighted sum of the per-stream auxiliary losses. Full hyperparameters are given in Appendix A.

Chapter 4

Results and Discussions

4.1 In-Domain Baseline

On the 140k test set (same StyleGAN fake distribution as training), the ViT baseline performs almost perfectly, as shown in Table 4.1. Such a high score is expected for a single-generator benchmark: the model only needs to learn one generator’s fingerprint.

Metric	Value
Accuracy	0.9962
Precision	0.9941
Recall	0.9984
F1	0.9963
ROC-AUC	1.000

Table 4.1: In-domain performance of the ViT baseline on the 140k (StyleGAN) test set.

4.2 Cross-Dataset Generalization

When the same baseline is tested on DFFD, which contains fake types it never saw, performance drops sharply (the “before” column of Table 4.2). In particular, recall falls to about 0.29: the model passes most non-StyleGAN fakes through as real, even though its precision stays high. After fine-tuning on DFFD’s diverse fakes, all metrics improve markedly (the “after” column), with recall rising to about 0.79.

Metric	Before	After	Δ
Accuracy	0.6465	0.8940	+0.2475
Precision	0.9966	0.9937	−0.0029
Recall	0.2940	0.7930	+0.4990
F1	0.4541	0.8821	+0.4280
ROC-AUC	0.9191	0.9635	+0.0444

Table 4.2: DFFD test-set performance before and after fine-tuning on diverse fakes.

Table 4.3 breaks the detection rate down by fake type. Entire-synthesis fakes (PGGAN, StyleGAN) become almost perfectly detectable after fine-tuning, and StarGAN improves substantially. FaceApp, however, remains the hardest: its subtle, localised attribute edits change only part of a real face and leave the weakest global artifacts, so it benefits least from fine-tuning. This directly motivates the frequency and noise-residual streams, which target exactly these subtle local cues.

4.3 Multi-Stream Fusion Results

The three-stream model was trained and evaluated on DFFD’s held-out, identity-disjoint test set of 53,381 images. It reaches a fake-detection recall of 0.9921, F1 of 0.9934 and ROC-AUC of 0.9988 (Table 4.4), while wrongly flagging only 2.6% of real faces. Fig. 4.1 shows the per-generator detection

Fake generator	Before	After	Δ
FaceApp (attribute edit)	0.380	0.452	+0.072
PGGAN (synthesis)	0.252	0.960	+0.708
StarGAN (attribute edit)	0.252	0.760	+0.508
StyleGAN-FFHQ (synthesis)	0.292	1.000	+0.708

Table 4.3: Per-generator detection rate (recall on each fake type) before and after fine-tuning the single-stream baseline on DFFD.

rate: the three synthesis / global-manipulation generators are effectively saturated, and FaceApp – the category that motivated the entire design – rises to 0.929.

Metric	Value
Accuracy	0.9891
Precision	0.9947
Recall (fake)	0.9921
F1	0.9934
ROC-AUC	0.9988

Table 4.4: Multi-stream fusion model on the DFFD held-out test set.

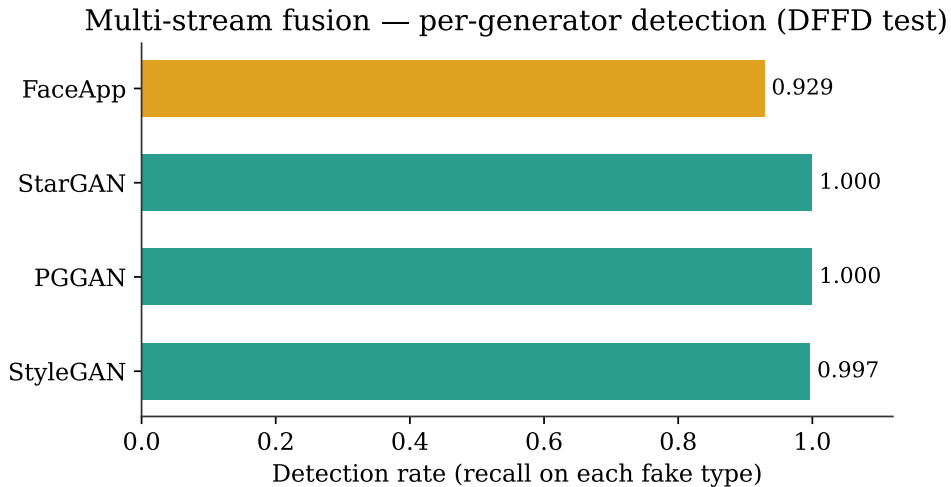


Figure 4.1: Per-generator detection rate of the multi-stream model on the DFFD test set. FaceApp remains the weakest category, but at 0.93 is far above the single-stream baseline’s 0.45.

The headline story is the recall trajectory in Fig. 4.2. Fake recall moves from 0.29 (single-stream ViT on unseen generators), to 0.79 (the same ViT fine-tuned on DFFD), to 0.99 once the frequency and noise-residual streams are fused in. FaceApp specifically improves from 0.45 under the fine-tuned single stream to 0.93 under fusion, confirming that complementary spectral and noise cues recover the local manipulations a semantic backbone misses.

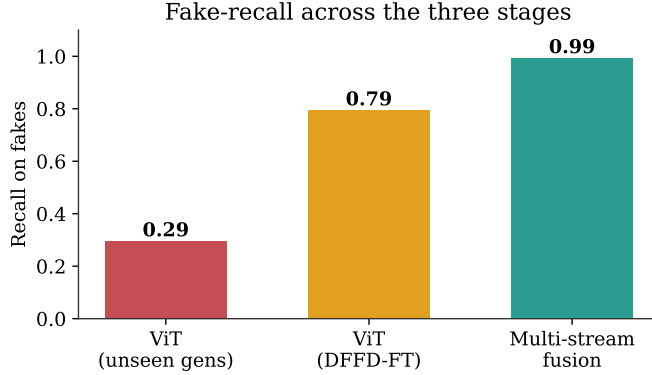


Figure 4.2: Fake-recall across the three stages of the project.

4.4 A Training Finding: Frozen Beats Fine-Tuned

A staged schedule that unfroze the RGB backbone for end-to-end fine-tuning was found to *degrade* the model rather than improve it. As shown in Fig. 4.3, the model’s best validation state occurred at the first epoch, while the backbone was frozen. After unfreezing, calibration collapsed: at one epoch the model flagged 46% of real faces as fake, and it never regained its earlier operating point. Keeping the pretrained backbone frozen throughout was both more accurate and roughly three times faster per epoch, and was adopted as the final recipe. The interpretation is that a confident, already-fine-tuned backbone does not need – and is destabilised by – joint fine-tuning alongside freshly initialised streams.

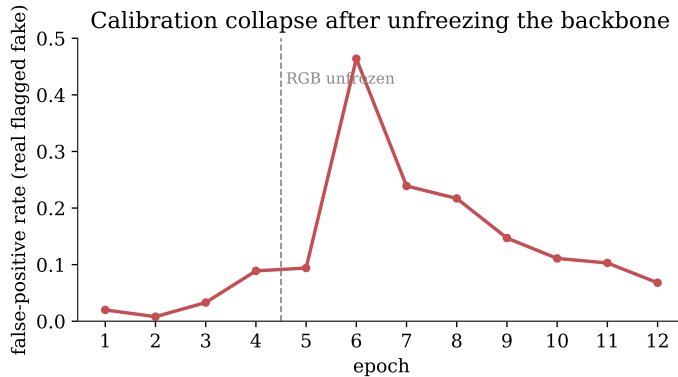


Figure 4.3: False-positive rate on real images by epoch. Unfreezing the RGB backbone (dashed line) destabilised calibration.

4.5 Discussion

The results establish a clear narrative. A detector can look near-perfect in-domain (Table 4.1) yet fail badly on unseen fake types (Table 4.2): the dangerous failure mode is *missed fakes*, not false alarms, since precision stays high while recall collapses. Training on diverse data closes much of this gap, but the residual weakness on attribute-manipulation fakes such as FaceApp shows that a single RGB stream is not sufficient. Fusing complementary frequency and noise-residual evidence closes the remaining gap, taking fake recall to 0.99 and lifting the hardest category, FaceApp, to 0.93 on held-out data. Together these results justify – and validate – the multi-encoder, multi-modal design.

Chapter 5

Explainability and Deployment

Aggregate metrics do not tell an operator *why* a given image was flagged. For each image the model therefore exposes four artefacts: a saliency map over the fake probability (computed as the gradient of $P(\text{fake})$ with respect to the input), the FFT log-magnitude spectrum read by the frequency stream, the SRM residual read by the noise stream, and a per-stream confidence obtained from the auxiliary heads. The per-stream confidence is the most useful signal for attribution: on fully synthesised faces all three streams agree, whereas on a FaceApp-style local edit the RGB stream is often unconvinced while the noise-residual stream drives the decision. Fig. 5.1 shows two representative panels.

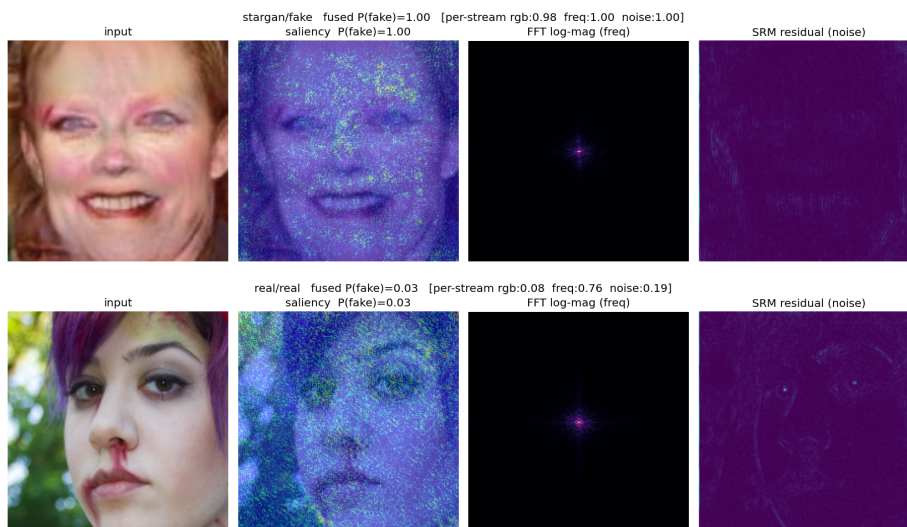


Figure 5.1: Explainability panels. For each image, left to right: input, saliency over $P(\text{fake})$, FFT log-magnitude (frequency stream), and SRM residual (noise stream); per-stream confidences are printed above each panel.

The trained detector is deployed behind a lightweight interactive interface built with Gradio. A user uploads a face image and receives the real/fake verdict with confidence, the three per-stream confidences shown as bars, and the saliency, FFT and SRM panels. The inference path deliberately mirrors the training pipeline – raw $[0, 1]$ tensors at 224×224 , with per-stream preprocessing applied internally – so that deployment behaviour matches evaluation. The full training, evaluation, explainability and interface code is maintained in the project repository (Appendix B).

Chapter 6

Conclusions and Future Work

This work established a ViT-based baseline for deepfake image detection that reaches near-perfect in-domain accuracy, and then used a controlled cross-dataset study on DFFD to demonstrate a severe generalization gap, with detection of unseen fake types dropping to roughly chance-level recall before recovering after fine-tuning. A per-generator analysis identified subtle attribute manipulations (FaceApp) as the hardest case. Guided by these findings, a three-stream detector combining RGB, frequency and noise-residual encoders through cross-attention fusion was implemented and trained end-to-end. On the held-out DFFD test set it achieves a fake recall of 0.992 and ROC-AUC of 0.999, raises FaceApp recall from 0.45 to 0.93, and is deployed behind an explainable verification interface. A secondary but practically important finding was that keeping the pretrained backbone frozen outperformed end-to-end fine-tuning, which destabilised calibration.

Future work will extend the fusion model to diffusion-generated faces (the pipeline is generator-agnostic and requires only additional labelled folders); move to video benchmarks such as Face-Forensics++ [11] and Celeb-DF [10] via frame extraction; test robustness under compression, blur and resizing; and increase the auxiliary-loss weight so the noise-residual stream carries still more of the FaceApp decision. A continual-learning strategy, so that the detector keeps pace with newly emerging generators rather than degrading over time, remains the longer-term goal.

Chapter 7

Recommendations

- Evaluate every detector across multiple generators and datasets, not a single benchmark, since in-domain accuracy alone is misleading.
- Train on diverse fake types (multiple GANs, diffusion and manipulation apps) to reduce the cross-generator generalization gap.
- Prioritise recall (detection rate) on unseen fakes as a key metric, because the typical failure is silently missing fakes.
- Combine complementary signals (RGB, frequency, noise residual) rather than relying on a single view, especially to catch subtle attribute edits.
- When fusing a strong pretrained backbone with freshly initialised streams, prefer freezing the backbone; joint end-to-end fine-tuning can destabilise calibration.
- Build explainability and continual adaptation into the system from the start so that it remains interpretable and current as generative techniques evolve.

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Appendix A

Training Configuration

Table A.1 lists the main settings for the baseline, the DFFD fine-tuning, and the multi-stream fusion runs.

Setting	Value
RGB backbone	ViT-Base/16 (ImageNet pre-trained, via <code>timm</code>)
Stream backbones	FFT-CNN and SRM+Bayar CNN (ResNet-18)
Fusion	Cross-attention transformer, 3 layers, fusion dim 512
Input size	224×224 , raw $[0, 1]$
Optimiser	AdamW (weight decay 10^{-4})
Learning rate	2×10^{-5} (baseline); 1×10^{-5} (fine-tune); 3×10^{-4} (fusion)
Loss	Cross-entropy + weighted per-stream auxiliary losses
Batch size / epochs	32 / 3 (baseline); 16 (eff. 32) / 4 (fusion)
Precision	Mixed precision (AMP); FFT/SRM computed in fp32
Hardware	Google Colab, NVIDIA T4 GPU

Table A.1: Training configuration.

Appendix B

Reproducibility

All experiments were run in Google Colab on a T4 GPU. The manifest builder reads DFFD’s official identity-disjoint split directly, so the train / validation / test partitions are reproducible. Model checkpoints (the StyleGAN baseline, the DFFD fine-tuned model, the warm-started frequency and noise streams, and the final fusion model), the training and evaluation scripts, the explainability tool and the interactive interface are maintained in the project repository. The label convention used throughout is fake = 1, real = 0, with fake treated as the positive class for recall, precision and ROC-AUC.